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United States Patent	12400606
Kind Code	B2
Date of Patent	August 26, 2025
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Display driving apparatus and method

Abstract

A display driving apparatus includes a display panel comprising a first display area displaying a first image and a second display area being maintained in a turn-off state during a hybrid mode; a source driving circuit configured to supply a first source signal to first pixels included in the first display area and supply a second source signal to second pixels included in the second display area to sense electrical characteristics of the second pixels when the display panel operates in the hybrid mode; and a timing controller configured to transfer first image data for the first image and sensing data for sensing to the source driving circuit and compensate for second image data for a second image which is to be displayed after displaying of the first image, based on a target compensation value obtained from the electrical characteristics sensed by the source driving circuit, wherein the target compensation value is determined based on a third image displayed in a previous display mode of the hybrid mode.

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Appl. No.: 18/733028

Filed: June 04, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240321221 A1	Sep. 26, 2024

Foreign Application Priority Data

KR	10-2021-0191102	Dec. 29, 2021
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Related U.S. Application Data

continuation parent-doc US 18063200 20221208 US 12033587 child-doc US 18733028

Publication Classification

Int. Cl.: G09G3/3275 (20160101); B60K35/00 (20240101); B60K35/22 (20240101); B60K35/28 (20240101); G09G3/3233 (20160101); G09G3/3266 (20160101)

U.S. Cl.:

CPC G09G3/3275 (20130101); B60K35/00 (20130101); G09G3/3233 (20130101); G09G3/3266 (20130101); B60K35/22 (20240101); B60K35/28 (20240101); B60K2360/16 (20240101); G09G2300/0842 (20130101); G09G2310/0213 (20130101); G09G2310/08 (20130101); G09G2320/0233 (20130101); G09G2320/0295 (20130101); G09G2320/045 (20130101); G09G2320/0613 (20130101); G09G2360/16 (20130101); G09G2380/10 (20130101)

Field of Classification Search

CPC: G09G (3/3233); G09G (3/3266); G09G (3/3275); G09G (2320/0285); G09G (2310/08); G09G (2320/0686); G09G (2310/0213); G09G (2320/0233); G09G (2300/0842); G09G (2310/04); G09G (2360/16); G09G (2320/0613); G09G (2380/10); G09G (2320/045); B60K (35/00); B60K (2370/16); B60K (2370/152)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/063,200, filed on Dec. 8, 2022, which claims the priority of the

BACKGROUND

Field of the Disclosure

(1) The present disclosure relates to a display apparatus, and more particularly, to a source drive integrated circuit (IC) for sensing a characteristic of a driving transistor.

DESCRIPTION OF THE BACKGROUND

(2) Organic light emitting display apparatuses of an active matrix type include an organic light emitting diode (OLED) which self-emits light and have advantages where a response time is fast, emission efficiency and luminance are high, and a viewing angle is wide.

(3) OLEDs which are self-emitting devices include an anode electrode, a cathode electrode, and an organic compound layer (for example, HIL, HTL, EML, ETL, and EIL) formed therebetween. The organic compound layer includes a hole injection layer (HIL), a hole transport layer (HTL), an emission layer (EML), an electron transport layer (ETL), and an electron injection layer (EIL). When a driving voltage is applied to the anode electrode and the cathode electrode, a hole passing through the hole transport layer (HTL) and an electron passing through the electron transport layer (ETL) move to the emission layer (EML) to generate an exciton, and thus, visible light is emitted from the emission layer (EML).

(4) In the organic light emitting display apparatuses, pixels including an OLED are arranged as a matrix type, and luminance of the pixels is adjusted based on a gray level of video data. Each of the pixels includes a driving transistor (for example, a driving thin film transistor (TFT)) which controls a driving current flowing in the OLED, based on a voltage V_{gs} between a gate electrode and a source electrode thereof. An electrical characteristic, such as a threshold voltage and mobility, of the driving transistor may be degraded as a driving time elapses, and due to this, a deviation may occur in each pixel. When the electrical characteristic of the driving transistor is changed for each pixel, luminance between pixels corresponding to the same video data is changed, and due to this, it is difficult to implement a desired image.

(5) Therefore, technology for sensing a characteristic value of the driving transistor and compensating for a deviation between characteristic values of driving transistors has been developed. However, it is difficult to intactly apply conventional compensation technology to various types of organic light emitting display apparatuses such as foldable display apparatuses, rollable display apparatuses, stretchable display apparatuses, or digital cockpit, or even when the conventional compensation technology is applicable, there is a problem where efficiency is reduced.

SUMMARY

(6) Accordingly, the present disclosure is to provide a display driving apparatus and a display driving method that substantially obviate one or more problems due to limitations and disadvantages described above.

(7) More specifically, the present disclosure is to provide a display driving apparatus and a display driving method, which may sense a characteristic of a driving transistor when a display panel is driven as a turn-on region where an image is displayed and a turn-off region where an image is not displayed.

(8) The present disclosure is also to provide a display driving apparatus and a display driving method, which may obtain a compensation value corresponding to a full region of a display panel based on a sensing result of a turn-off region.

(9) Further, the present disclosure is to provide a display driving apparatus and a display driving method, which may determine an optimal compensation value based on a content type of an image displayed through a full region of a display panel.

(10) Additional advantages and features of the disclosure will be set forth in part in the description

which follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or may be learned from practice of the disclosure. Other advantages of the disclosure may be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

(11) To achieve these and other advantages and in accordance with the present disclosure, as embodied and broadly described herein, a display driving apparatus includes a display panel comprising a first display area displaying a first image and a second display area being maintained in a turn-off state during a hybrid mode; a source driving circuit configured to supply a first source signal to first pixels included in the first display area and supply a second source signal to second pixels included in the second display area to sense electrical characteristics of the second pixels when the display panel operates in the hybrid mode; and a timing controller configured to transfer first image data for the first image and sensing data for sensing to the source driving circuit and compensate for second image data for a second image which is to be displayed after displaying of the first image, based on a target compensation value obtained from the electrical characteristics sensed by the source driving circuit, wherein the target compensation value is determined based on a third image displayed in a previous display mode of the hybrid mode.

(12) In another aspect of the present disclosure, a vehicle includes a display panel including a first display area to display a first image and a second display area being maintained in a turn-off state during a hybrid mode; a source driving circuit configured to supply a first source signal to first pixels disposed in the first display area and supply a second source signal to second pixels disposed in the second display area to sense electrical characteristics of the second pixels when the display panel operates in the hybrid mode; and a timing controller configured to transfer first image data for the first image and sensing data for sensing to the source driving circuit and compensate for second image data for a second image to be displayed after displaying of the first image, based on a target compensation value obtained from the electrical characteristics sensed by the source driving circuit, wherein the target compensation value is determined based on a third image displayed in a previous display mode of the hybrid mode, and when a vehicle driving start signal and a display panel raising signal are received from an electronic control unit (ECU) of the vehicle, the timing controller sets an operation mode of the display apparatus to the hybrid mode.

(13) It is to be understood that both the foregoing general description and the following detailed description of the present disclosure are exemplary and explanatory and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate aspects of the disclosure and together with the description serve to explain the principle of the disclosure.

(2) In the drawings:

(3) FIG. 1 is a diagram illustrating a display apparatus according to an aspect of the present disclosure;

(4) FIGS. 2A, 2B, 2C and 2D are diagrams illustrating an example of the state of a display panel based on an operation mode of a display apparatus;

(5) FIG. 3 is a diagram illustrating a pixel configuration according to an aspect of the present disclosure;

(6) FIG. 4 is a block diagram illustrating a configuration of a timing controller according to an aspect of the present disclosure;

(7) FIGS. 5A, 5B, 5C and 5D are diagrams illustrating a turn-off region and a turn-on region of a

display panel and an operation of gate lines in a hybrid mode operation according to an aspect of the present disclosure; and

(8) FIG. 6 is a diagram schematically illustrating a configuration of each of a source drive IC and a display panel according to an aspect of the present disclosure.

DETAILED DESCRIPTION

(9) In the disclosure, it should be noted that like reference numerals already used to denote like elements in other drawings are used for elements wherever possible. In the following description, when a function and a configuration known to those skilled in the art are irrelevant to the essential configuration of the present disclosure, their detailed descriptions will be omitted. The terms described in the disclosure should be understood as follows.

(10) Advantages and features of the present disclosure, and implementation methods thereof will be clarified through following aspects described with reference to the accompanying drawings. The present disclosure may, however, be embodied in different forms and should not be construed as limited to the aspects set forth herein. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art. Further, the present disclosure is only defined by scopes of claims.

(11) A shape, a size, a ratio, an angle, and a number disclosed in the drawings for describing aspects of the present disclosure are merely an example, and thus, the present disclosure is not limited to the illustrated details. Like reference numerals refer to like elements throughout. In the following description, when the detailed description of the relevant known function or configuration is determined to unnecessarily obscure the important point of the present disclosure, the detailed description will be omitted.

(12) In a case where ‘comprise’, ‘have’, and ‘include’ described in the present disclosure are used, another part may be added unless ‘only’ is used. The terms of a singular form may include plural forms unless referred to the contrary.

(13) In construing an element, the element is construed as including an error range although there is no explicit description.

(14) In describing a time relationship, for example, when the temporal order is described as ‘after’, ‘subsequent’, ‘next’, and ‘before’, a case which is not continuous may be included unless ‘just’ or ‘direct’ is used.

(15) It will be understood that, although the terms “first”, “second”, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element without departing from the scope of the present disclosure.

(16) The term “at least one” should be understood as including any and all combinations of one or more of the associated listed items. For example, the meaning of “at least one of a first item, a second item, and a third item” denotes the combination of all items proposed from two or more of the first item, the second item, and the third item as well as the first item, the second item, or the third item.

(17) Features of various aspects of the present disclosure may be partially or overall coupled to or combined with each other, and may be variously inter-operated with each other and driven technically as those skilled in the art may sufficiently understand. The aspects of the present disclosure may be carried out independently from each other, or may be carried out together in co-dependent relationship.

(18) Hereinafter, aspects of the present disclosure will be described in detail with reference to the accompanying drawings.

(19) FIG. 1 is a diagram illustrating a display apparatus **100** according to an aspect of the present disclosure. Referring to FIG. 1, the display apparatus **100** according to an aspect of the present disclosure may include a display panel **10**, a timing controller **11**, a source driving circuit **12**, and a gate driving circuit **13**. The timing controller **11**, the source driving circuit **12**, and the gate driving

circuit **13** may configure a display driving apparatus.

(20) The display panel **10** may display image data RGB' input from the timing controller **11**. In an aspect, the display panel **10** may be a display panel for foldable display apparatuses, rollable display apparatuses, stretchable display apparatuses, or digital cockpit.

(21) According to such an aspect, the display panel **10** may operate in one of a display mode, a sensing mode, and a hybrid mode, based on an operation mode of the display driving apparatus. The display mode may denote a mode where an image is displayed on a full region of the display panel **10**. The sensing mode may denote a mode where sensing is performed through the full region of the display panel **10**. The hybrid mode may denote a mode where an image is display on a first display area which is a portion of the display panel **10** and sensing is performed through a second display area which is the other region except the first display area. When the display panel **10** operates in the sensing mode, the display panel **10** may be maintained in a turn-off state where an image is not displayed. Also, when the display panel **10** operates in the hybrid mode, the second display area of the display panel **10** may be maintained in the turn-off state.

(22) In an aspect, in a case where the display panel **10** is implemented as a digital cockpit, the display panel **10** may operate in the display mode when a vehicle including the display panel **10** is not driving, and thus, an image may be displayed on a full region of the display panel **10**.

(23) Moreover, the display panel **10** may operate in the sensing mode during a power-off period of the display driving apparatus, and thus, sensing may be performed through the display panel **10**.

(24) In the aspect described above, it has been described that the sensing mode is performed during the power-off period of the display driving apparatus, but the present disclosure is not limited thereto and the sensing mode may be performed during a blank period where an image is not displayed in performing an operation of the display mode. In this case, sensing may be performed in a partial region, instead of a full region, of the display panel **10**.

(25) Moreover, the display panel **10** may operate in the hybrid mode when the vehicle is driving, and thus, an image for an instrument panel may be displayed on the first display area and the second display area may be maintained in a turn-off state where an image is not displayed.

(26) For example, when the vehicle is not driving, as illustrated in FIG. 2A, a full region of the display panel **10** may be raised from the inside of a dashboard **200** to the outside and may be exposed at the outside. Thus, the display panel **10** may operate in the display mode where an image is displayed on the full region.

(27) On the other hand, when the display panel **10** is turned off due to the power-off of the display apparatus as illustrated in FIG. 2B or the full region of the display panel **10** is maintained in the dashboard **200** as illustrated in FIG. 2C, the display panel **10** may operate in the sensing mode.

(28) Moreover, as illustrated in FIG. 2D, when the vehicle is driving, a first display area **10a** of the display panel **10** may be raised from the inside of the dashboard **200** to the outside and may be exposed at the outside. Thus, an image for an instrument panel may be displayed on the first display area **10a** and a second display area **10b** of the display panel **10** may be maintained in a turn-off state where an image is not displayed in the dashboard **200**.

(29) As illustrated in FIG. 1, a plurality of data lines **14A**, a plurality of sensing lines **14B**, and a plurality of gate lines **15** may be formed in the display panel **10** described above. The plurality of data lines **14A** and the plurality of sensing lines **14B** may be formed to intersect with the plurality of gate lines **15**.

(30) A plurality of pixels P may be arranged as a matrix type in a plurality of areas defined by intersections of the plurality of data lines **14A** and the plurality of gate lines **15**. In an aspect, each of the pixels P may include an organic light emitting diode (OLED) (not shown) and one or more thin film transistors (TFTs) (not shown).

(31) The pixel P may be connected to one of the data lines **14A**, one of the sensing lines **14B**, and one of the gate lines **15**. In response to a gate pulse input through the gate line **15**, the pixel P may receive a source signal through the data line **14A** and may output characteristic information about a

driving transistor (not shown) included in each pixel P through the sensing line **14B**. In an aspect, the characteristic information about the driving transistor may include a threshold voltage or mobility of the driving transistor.

(32) The pixel P may be supplied with a high level driving voltage EVDD and a low level driving voltage EVSS from a power generator (not shown).

(33) An example of a configuration of the pixel P is illustrated in FIG. 3. Referring to FIG. 3, the pixel P according to the present disclosure may include a first switching transistor ST1, a second switching transistor ST2, a storage capacitor Cst, a driving transistor DT, and an OLED.

(34) The first switching transistor ST1 may apply a source signal (Vdata-D or Vdata-S) transferred through the data line **14A** to a first node N1 in response to a gate pulse SCAN. The first switching transistor ST1 may include a gate electrode connected to the gate line **15**, a drain electrode connected to the data line **14A**, and a source electrode connected to the first node N1.

(35) The second switching transistor ST2 may switch a current flow between a second node N2 and the sensing line **14B** in response to the gate pulse SCAN. The second switching transistor ST2 may include a gate electrode connected to the gate line **15**, a drain electrode connected to the sensing line **14B**, and a source electrode connected to the second node N2.

(36) The storage capacitor Cst may be connected between the first node N1 and the second node N2.

(37) The driving transistor DT may control the amount of current input to the OLED based on a gate-source voltage Vgs thereof. The driving transistor DT may include a gate electrode connected to the first node N1, a drain electrode connected to an input terminal for the high level driving voltage EVDD, and a source electrode connected to the second node N2.

(38) The OLED may include an anode electrode connected to the second node N2, a cathode electrode connected to an input terminal for the low level driving voltage EVSS, and an organic compound layer disposed between the anode electrode and the cathode electrode.

(39) A pixel structure illustrated in FIG. 3 is merely an aspect for helping understand and may be variously modified, and thus, the inventive concept is not limited to the aspect.

(40) Referring again to FIG. 1, the timing controller **11** may control an operation of each of the source driving circuit **12** and the gate driving circuit **13**. In detail, the timing controller **11** may generate a source control signal SCS for controlling an operation timing of the source driving circuit **12** and a gate control signal GCS for controlling an operation timing of the gate driving circuit **13**, based on timing signals TS such as a vertical synchronization signal Vsync, a horizontal synchronization signal Hsync, a dot clock signal DCLK, and a source output enable signal SOE.

(41) In an aspect, the timing controller **11** may set an operation mode of the display apparatus **100** and may control an operation of each of the source driving circuit **12** and the gate driving circuit **13** based on the set operation mode. Particularly, the timing controller **11** according to the present disclosure may determine an operation mode of the display apparatus **100** as one of the display mode, the sensing mode, and the hybrid mode and may control an operation of each of the source driving circuit **12** and the gate driving circuit **13** so that the display apparatus **100** is driven, based on the determined operation mode.

(42) Hereinafter, a configuration of the timing controller **11** according to the present disclosure will be described in more detail with reference to FIG. 4.

(43) FIG. 4 is a block diagram illustrating a configuration of a timing controller according to an aspect of the present disclosure. As illustrated in FIG. 4, the timing controller **11** may include an operation mode setting unit **310**, a control signal generator **320**, a data processor **330**, a content type determiner **340**, and a data compensator **350**.

(44) The operation mode setting unit **310** may set an operation mode of the display apparatus **100**. In detail, the operation mode setting unit **310** may determine an operation mode of the display apparatus **100** as one of the display mode, the sensing mode, and the hybrid mode. The display mode is a mode where an image is displayed on a full region of the display panel **10**. The sensing

mode is a mode where a sensing value is obtained by sensing characteristic information about a driving transistor. The hybrid mode is a mode where an image is display on the first display area **10a** of the display panel **10** and the sensing value is obtained by sensing the characteristic information about the driving transistor through the second display area **10b**. In this case, the sensing mode may be performed in performing a power-on operation or a power-off operation, or may be performed in blank periods while the display mode is being driven.

(45) The operation mode setting unit **310** may determine an operation mode of the display apparatus **100** based on a vehicle driving-related signal CDS, a display panel elevation signal DMS, etc. In this case, the vehicle driving-related signal CDS and the display panel elevation signal DMS may be generated by an electronic control unit (ECU) of the vehicle and may be input to the operation mode setting unit **310**.

(46) For example, when a display panel raising signal and a vehicle driving start signal are received from the ECU of the vehicle, the operation mode setting unit **310** may set an operation mode of the display apparatus **100** to the hybrid mode.

(47) When a display panel lowering signal or a power-off signal of the display apparatus off signal received from the ECU of the vehicle, the operation mode setting unit **310** may set an operation mode of the display apparatus to the sensing mode.

(48) When the display panel raising signal and a vehicle driving stop signal received from the ECU of the vehicle, the operation mode setting unit **310** may set an operation mode of the display apparatus **100** to the display mode.

(49) In an aspect, when an operation mode of the display apparatus **100** is determined as the hybrid mode, the operation mode setting unit **310** may generate information about target gate lines where sensing is to be performed for each frame during the hybrid mode and may provide the generated information to the control signal generator **320**. In this case, the positions and number of target gate lines where sensing is to be performed may be set to vary by units of one frame or by units of two or more frames, or may be set so that a variation order of the target gate lines is repeated by units of a predetermined number of frames.

(50) The control signal generator **320** may generate a source control signal SCS for controlling an operation timing of the source driving circuit **12** and a gate control signal GCS for controlling an operation timing of the gate driving circuit **13** so that the source driving circuit **12** and the gate driving circuit **13** operate in a set operation mode, based on timing signals TS supplied from the outside and an operation mode setting signal MS supplied from the operation mode setting unit **310**. The control signal generator **320** may transfer the generated source control signal SCS to the source driving circuit **12** and may transfer the gate control signal GCS to the gate driving circuit **13**.

(51) Particularly, the control signal generator **320** may generate the gate control signal GCS based on information about a target gate line set by the operation mode setting unit **310** when the display apparatus **100** operates in the hybrid mode.

(52) When the operation mode setting signal MS supplied from the operation mode setting unit **310** indicates the display mode, the data processor **330** may convert video data RGB, input from the outside, into image data RGB' capable of being processed by the source driving circuit **12** and may provide the image data RGB' to the source driving circuit **12**.

(53) When the operation mode setting signal MS supplied from the operation mode setting unit **310** indicates the sensing mode, the data processor **330** may generate sensing data SDC for sensing characteristic information about a driving transistor of a pixel P which is to be sensed and may provide the sensing data SDC to the source driving circuit **12**.

(54) When the operation mode setting signal MS supplied from the operation mode setting unit **310** indicates the hybrid mode, the data processor **330** may generate image data RGB' which are to be supplied to pixels P included in the first display area **10a** and sensing data SDC which is to be supplied to pixels P included in the second display area **10b** and may provide the image data RGB' and the sensing data SDC to the source driving circuit **12**.

(55) The content type determiner **340** may determine a content type of an image displayed on a full region of the display panel **10** before the display apparatus **100** operates in the hybrid mode based on a sensing result RSD transferred from the source driving circuit **12** when the display apparatus **100** is driven in the hybrid mode.

(56) In an aspect, the content type may include a normal type which includes general images such as movie or drama, a graphics type which includes images such as a user interface (UI) or a vehicular cluster, a sports type which includes a sports image, and a game type which includes a game image.

(57) In the present disclosure, the reason that the content type determiner **340** determines a content type of an image displayed on the full region of the display panel **10** is following. When the display apparatus **100** operates in the hybrid mode, sensing values for the full region of the display panel **10** may not be obtained because sensing is performed in only the second display area **10b** which is a portion of the display panel **10**. Thus, the content type determiner **340** determines the content type of the image displayed on the full region of the display panel **10** and obtains a compensation value corresponding to the full region of the display panel **10** mapped to corresponding content type.

(58) In an aspect, the content type determiner **340** may determine a content type of an image displayed on the full region of the display panel **10** in the display mode which is performed before the hybrid mode by inputting the sensing result RSD to a first artificial neural network model **342**. The sensing result RSD may be transferred from the source driving circuit **12**.

(59) In this case, because the first artificial neural network model **342** is previously trained as a training set including a sensing result which is obtained in outputting image data corresponding to each content type for each content type. Thus, when a specific sensing result is input, the first artificial neural network model **342** may return a value of a probability that the input sensing result is a corresponding content type for each content type.

(60) In an aspect, the first artificial neural network model **342** may be a convolution neural network (CNN)-based algorithm or feature-based algorithm.

(61) In FIG. **4**, the content type determiner **340** is illustrated as being included in the timing controller **11**, but this is merely an aspect and a function performed by the content type determiner **340** may be performed by a micro controller unit (MCU) or a microprocessor unit (MPU) of the display apparatus **100**. According to such an aspect, the source driving circuit **12** may provide the MCU or the MPU with the sensing result RSD which is obtained in the hybrid mode, and the MCU or the MPU may determine a content type based on the sensing result RSD and may provide the determined content type to the data compensator **350**.

(62) The data compensator **350** may obtain a compensation value, mapped to the content type determined by the content type determiner **340**, as a target compensation value. In an aspect, each content type-based compensation value may be stored in a memory (not shown). In another aspect, each content type-based compensation value may be obtained by a second artificial neural network model (not shown). When sensing results obtained in displaying an image corresponding to each content type on a full screen are input, the second artificial neural network model may be previously trained to output a compensation value corresponding thereto. Therefore, each content type-based compensation value may be obtained by inputting a content type, determined by the content type determiner **340**, to the second artificial neural network model. To this end, the data compensator **350** may further include the second artificial neural network model. In another aspect, when the MCU or the MPU determines a content type based on the sensing result RSD, the MCU or the MPU may obtain a target compensation value based on the determined content type and may provide the target compensation value to the data compensator **350**.

(63) When the display mode is again driven after the target compensation value is obtained and the hybrid mode is driven, the data compensator **350** may modulate image data which is to be displayed on the full region of the display panel **10** in driving of the display mode. The data compensator **350** may modulate the image data by adding or multiplying the target compensation

value to the image data.

(64) Furthermore, when the sensing result RSD obtained through an operation of the sensing mode is received from the source driving circuit **12**, the data compensator **350** may calculate a compensation value for compensating for an electrical characteristic change of a driving transistor included in each pixel P by using a predetermined compensation algorithm. The data compensator **350** may modulate image data which is to be provided to the source driving circuit **12** based on the calculated compensation value. In detail, the data compensator **350** may apply the sensing result RSD, transferred from the source driving circuit **12** in driving of the sensing mode, to a pre-stored compensation algorithm to calculate a threshold voltage deviation and a mobility change amount of the driving transistor included in each pixel P, and then, may calculate a compensation value for compensating for the change amount. The calculated compensation value may be stored in a compensation value storage unit (not shown). Whenever a new compensation value is calculated, a compensation value stored in the compensation value storage unit may be updated to the new compensation value.

(65) In an aspect, when a first image is displayed on the first display area **10a** during the hybrid mode and image data of a second image to be displayed on the full region of the display panel **10** in the display mode performed after the hybrid mode is compensated for, the data compensator **350** may compare luminance of the first image with luminance of a third image displayed in the display mode performed before the hybrid mode to determine a compensation value which is to be used to compensate for the second image.

(66) In detail, when luminance of the first image is lower than luminance of the third image, the data compensator **350** may compensate for the image data of the second image by using a target compensation value obtained based on a sensing result obtained in the hybrid mode.

(67) On the other hand, when luminance of the first image is higher than luminance of the third image, the data compensator **350** may compensate for the image data of the second image by using a compensation value obtained based on a sensing result obtained through the sensing mode performed after an operation of the hybrid mode. This may be because much distortion may occur in image data due to an inaccurate compensation value in a case where the image data of the second image is compensated for based on a compensation value determined based on a content type of the third image, when luminance of the first image is higher than luminance of the third image.

(68) Referring again to FIG. **1**, in driving of the display mode, the gate driving circuit **13** may generate a gate pulse for displaying an image based on the gate controls signal GCS, and then, may sequentially supply the generated gate pulse to the gate lines **15**. Accordingly, all gate lines **15** may be turned on by a row progressive driving scheme, and thus, an image may be displayed on the full region of the display panel **10**.

(69) In driving of the sensing mode, the gate driving circuit **13** may generate a gate pulse for sensing of each pixel P based on the gate control signal GCS, and then, may sequentially supply the generated gate pulse to each of the gate lines **15**. Accordingly, electrical characteristics of driving transistors included in pixels connected to each gate line **15** may be sequentially sensed.

(70) In driving of the hybrid mode, the gate driving circuit **13** may generate a gate pulse for displaying of an image on the first display area **10a** based on the gate control signal GCS to sequentially supply the generated gate pulse to the first gate lines included in the first display area **10a**. In driving of the hybrid mode, the gate driving circuit **13** may generate a gate pulse for sensing of pixels connected to target gate lines among second gate lines included in the second display area **10b** to sequentially supply the generated gate pulse to the target gate lines. In this case, the target gate line may be configured with two or more gate lines and may be changed based on a predetermined frame period.

(71) For example, as illustrated in FIGS. **5A** to **5D**, when the display apparatus **100** is being driven in the hybrid mode, first gate lines **500a** to **500n** included in the first display area **10a** may be sequentially turned on at every frame in N_{th} to $(N+3)_{th}$ frames, and thus, an image may be

displayed through pixels of the first display area **10a**.

(72) Furthermore, as illustrated in FIG. 5A, in the N.sup.th frame, only first target gate lines **510a** and **510b** may be turned on in the second display area **10b**, and thus, electrical characteristics of pixels connected to the first target gate lines **510a** and **510b** may be sensed. Also, as illustrated in FIG. 5B, in the N+1.sup.th frame, only second target gate lines **520a** and **520b** may be turned on in the second display area **10b**, and thus, electrical characteristics of pixels connected to the second target gate lines **520a** and **520b** may be sensed.

(73) Moreover, as illustrated in FIG. 5C, in the N+2.sup.th frame, only third target gate lines **530a** and **530b** may be turned on in the second display area **10b**, and thus, electrical characteristics of pixels connected to the third target gate lines **530a** and **530b** may be sensed. Also, as illustrated in FIG. 5D, in the N+3.sup.th frame, only fourth target gate lines **540a** and **540b** may be turned on in the second display area **10b**, and thus, electrical characteristics of pixels connected to the fourth target gate lines **540a** and **540b** may be sensed.

(74) In FIGS. 5A to 5D, it has been described that two target gate lines are turned on in the second display area **10b** at every frame, but this is merely an aspect and three or more target gate lines may be selected. Also, in FIGS. 5A to 5D, it has been described that a target gate line is changed by units of one frame, but a target gate line may be changed by units of two or more frames or a change order of target gate lines may be repeated by units of a predetermined number of frames.

(75) In the aspects described above, a gate pulse applied to gate lines for sensing in the hybrid mode or the sensing mode may be greater in on pulse period than a gate pulse applied to gate lines for displaying of an image in the display mode or the hybrid mode. An on pulse period of a gate pulse used for sensing may correspond to a one line sensing on time. Here, the one line sensing on time may denote a scan time needed for simultaneously sensing pixels P included in one horizontal line L#1 to L#p.

(76) Referring again to FIG. 1, the source driving circuit **12** may be connected to the plurality of data lines **14A** and may operate in the display mode, the sensing mode, and the hybrid mode based on mode control by the timing controller **11**. As illustrated in FIG. 1, the source driving circuit **12** may include a plurality of source drive ICs SDIC#1 to SDIC#k.

(77) In operating in the display mode, the source drive ICs SDIC#1 to SDIC#k may convert image data RGB' supplied from the timing controller **11** into a display source signal Vdata_D. The source drive ICs SDIC#1 to SDIC#k may supply the display source signal Vdata_D to each pixel P through a corresponding data line **14A**.

(78) In operating in the sensing mode, the source drive ICs SDIC#1 to SDIC#k may convert sensing data SDC supplied from the timing controller **11** into a sensing source signal Vdata_S. The source drive ICs SDIC#1 to SDIC#k may supply the sensing source signal Vdata_S to each pixel P through a corresponding data line **14A** to sense a characteristic of a driving transistor included in each pixel P. The source drive ICs SDIC#1 to SDIC#k may transfer a sensing result RSD obtained through sensing to the timing controller **11**.

(79) In operating in the hybrid mode, the source drive ICs SDIC#1 to SDIC#k may convert image data RGB' supplied from the timing controller **11** into the display source signal Vdata_D. The source drive ICs SDIC#1 to SDIC#k may supply the display source signal Vdata_D to each pixel P included in the first display area **10a** through a corresponding data line **14A**. Moreover, the source drive ICs SDIC#1 to SDIC#k may convert the sensing data SDC supplied from the timing controller **11** into the sensing source signal Vdata_S. The source drive ICs SDIC#1 to SDIC#k may supply the sensing source signal Vdata_S to each pixel P included in the second display area **10b** to corresponding data lines **14A** to sense a characteristic of the driving transistor included in each pixel P. The source drive ICs SDIC#1 to SDIC#k may transfer the sensing result RSD obtained through sensing in the second display area to the timing controller **11**.

(80) Hereinafter, a configuration of each of the source drive ICs SDIC#1 to SDIC#k according to the present disclosure will be described in more detail with reference to FIG. 6.

(81) FIG. 6 is a diagram schematically illustrating a configuration of each of a source drive IC and a display panel according to an aspect of the present disclosure. Configurations of source drive ICs SDIC#1 to SDIC#k may be the same, and thus, in FIG. 6, only a configuration of one source drive IC SDIC of a plurality of source drive ICs SDIC#1 to SDIC#k will be described for convenience of description.

(82) As illustrated in FIG. 6, a source drive IC SDIC may include a plurality of source signal input units **610** and a sensing block **620**.

(83) Each of the plurality of source signal input units **610** may be connected to a data line **14A** corresponding thereto and may input a display source signal Vdata_D or a sensing source signal Vdata_S to each pixel P through the corresponding data line **14A**. In an aspect, although not shown, to supply the display source signal Vdata_D or the sensing source signal Vdata_S to each pixel P, each of the source signal input units **610** may include elements such as a shift register, a latch, a digital-to-analog converter, and an output buffer.

(84) In driving of the display mode, the source signal input unit **610** may receive image data from the timing controller **11**, convert the received image data into the display source signal Vdata_D based on a source control signal SCS, and supply the display source signal Vdata_D to each data line **14A**.

(85) Moreover, in driving of the sensing mode, the source signal input unit **610** may receive sensing data from the timing controller **11**, convert the received sensing data into the sensing source signal Vdata_S based on the source control signal SCS, and supply the sensing source signal Vdata_S to the data lines **14A**.

(86) Moreover, in driving of the hybrid mode, the source signal input unit **610** may convert image data, received from the timing controller **11**, into the display source signal Vdata_D based on the source control signal SCS and may supply the display source signal Vdata_D to pixels P connected to first gate lines **15a** to **15n** of a first display area **10a**. Moreover, the source signal input unit **610** may convert sensing data, received from the timing controller **11**, into the sensing source signal Vdata_S for sensing based on the source control signal SCS and may supply the sensing source signal Vdata_S to pixels P connected to second gate lines TG1 and TG2 of a second display area **10b**.

(87) The sensing block **620** may be connected to sensing lines **14B** through sensing channels CH#1 to CH#n and may sense pixels P connected to each sensing line **14B**. In detail, the sensing block **620** may be connected to the sensing channels CH#1 to CH#n and may sense pixels P connected to corresponding sensing channels CH#1 to CH#n, thereby obtaining a sensing value. In detail, in the sensing mode or the hybrid mode, the sensing block **620** may obtain a sensing value from a driving transistor of a corresponding pixel P, based on the supply of a sensing source signal to each pixel P which is to be sensed.

(88) The sensing block **620** may convert the obtained sensing value into a digital value to generate a sensing result RSD and may transfer the generated sensing result RSD to the timing controller **11**. The sensing block **620** may include an analog-to-digital converter (not shown) for converting the obtained sensing value into a digital value.

(89) It may be understood that those skilled in the art may modify the present disclosure in other detailed forms without changing the technical spirit or the essential feature.

(90) For example, in the aspect described above, a sensing block is illustrated as being included in a source drive IC, but this is merely an aspect and the sensing block may be implemented as an element physically apart from the source drive IC.

(91) All disclosed methods and procedures described herein may be implemented, at least in part, using one or more computer programs or components. These components may be provided as a series of computer instructions through any conventional computer-readable medium or machine-readable medium including volatile and nonvolatile memories such as random-access memories (RAMs), read only-memories (ROMs), flash memories, magnetic or optical disks, optical

memories, or other storage media. The instructions may be provided as software or firmware, and may, in whole or in part, be implemented in a hardware configuration such as application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), digital signal processors (DSPs), or any other similar device. The instructions may be configured to be executed by one or more processors or other hardware configurations, and the processors or other hardware configurations are allowed to perform all or part of the methods and procedures disclosed herein when executing the series of computer instructions.

(92) According to the present disclosure, in a case where only a partial region of a display panel is displaying an image, an operation of sensing an electrical characteristic of a driving transistor and an operation of compensating for image data may be performed without waiting for the turn-off of the display panel, and thus, a sensing time of a display apparatus may be reduced, thereby increasing sensing efficiency.

(93) Moreover, according to the present disclosure, in a case where the display panel is driven as a turn-on region where an image is displayed and a turn-off region where an image is not displayed, characteristics of driving transistors included in the turn-off region may be sensed, and based thereon, a compensation value corresponding to a full region of the display panel may be predicted, whereby compensation data may be applied within an early time while the full region of the display panel displays an image again.

(94) Moreover, according to the present disclosure, by inputting characteristic sensing results of driving transistors included in the turn-off region of the display panel to an artificial neural network model, a content type of an image displayed on a full region of a corresponding display panel may be determined, and a compensation value corresponding to a full region of a corresponding display panel may be predicted based on the compensation value matching the determined content type, thereby calculating an optimal compensation value based on the type of content.

(95) Therefore, the above-described aspects should be understood to be exemplary and not limiting in every aspect. The scope of the present disclosure will be defined by the following claims rather than the above-detailed description, and all changes and modifications derived from the meaning and the scope of the claims and equivalents thereof should be understood as being included in the scope of the present disclosure.

Claims

1. A display apparatus comprising: a display panel including a first display area displaying a first image and a second display area being maintained in a turn-off state during a hybrid mode; a source driving circuit configured to supply a first source signal to first pixels disposed in the first display area, supply a second source signal to second pixels disposed in the second display area and sense electrical characteristics of the second pixels when the display panel operates in the hybrid mode; and a timing controller configured to transfer first image data for the first image and sensing data for sensing to the source driving circuit and compensate for second image data for a second image to be displayed after displaying of the first image, based on a target compensation value obtained from the electrical characteristics sensed by the source driving circuit, wherein the target compensation value is determined based on a third image displayed in a previous display mode of the hybrid mode.

2. The display apparatus of claim 1, wherein the timing controller comprises: a content type determiner configured to determine a content type of the third image displayed in the first display area and the second display area during the previous display mode; and a data compensator configured to determine the target compensation value based on a compensation value mapped to the content type and compensate for the second image data by using the target compensation value, wherein the second image data is image data for the second image to be displayed in the first display area and the second display area during a display mode to be performed after the hybrid

mode.

3. The display apparatus of claim 2, wherein luminance of the first image is lower than luminance of the third image.
4. The display apparatus of claim 2, wherein, when luminance of the first image is higher than luminance of the third image, the data compensator compensates for the second image data by using the compensation value determined based on a sensing result obtained through a sensing mode performed after an operation of the hybrid mode.
5. The display apparatus of claim 2, wherein the timing controller further comprises a first artificial neural network model trained using a training set including sensing values obtained in outputting image data corresponding to each content type for each content type, wherein the content type determiner inputs the sensing result to the first artificial neural network model to determine a result value, having a highest probability value among result values fed back from the first artificial neural network model, as a content type mapped to the sensing result.
6. The display apparatus of claim 2, wherein the content type comprises a normal type including images for movie or drama, a graphics type including images for a user interface (UI) or a vehicular cluster, a sports type including a sports image, and a game type including a game image.
7. The display apparatus of claim 1, further comprising a gate driving circuit configured to sequentially supply a gate pulse to first gate lines disposed in the first display area to drive the first gate lines and supply the gate pulse to more than one target gate lines among second gate lines disposed in the second display area to drive the target gate lines, wherein a plurality of continuous first gate lines are disposed in the first display area and a plurality of continuous second gate lines are disposed in the second display area.
8. The display apparatus of claim 7, wherein the target gate lines are changed at every frame period.
9. The display apparatus of claim 7, wherein, in operating in a display mode, the display panel displays an image in the first display area and the second display area, the source driving circuit supplies a source signal for displaying the image to the first and second pixels, and the gate driving circuit sequentially supplies the gate pulse to the gate lines disposed in the first and second display areas to sequentially drive the gate lines.
10. The display apparatus of claim 7, wherein, in operating in a sensing mode, the first display area and the second display area are maintained in a turn-off state, the source driving circuit supplies a source signal for sensing electrical characteristics of the first and second pixels to the first and second pixels, and the gate driving circuit sequentially supplies a gate pulse to gate lines included in the first display area and the second display area to sequentially drive the gate lines.
11. The display apparatus of claim 1, wherein the display panel is a digital cockpit equipped in a vehicle, when a vehicle driving start signal and a display panel raising signal are received from an electronic control unit (ECU) of the vehicle, the timing controller sets an operation mode of the display apparatus to the hybrid mode, when a display panel lowering signal or a display panel off signal is received from the ECU, the timing controller sets an operation mode of the display apparatus to a sensing mode, and when a vehicle driving stop signal and the display panel raising signal are received from the ECU, the timing controller sets an operation mode of the display apparatus to a display mode.
12. A vehicle comprising: a display panel including a first display area to display a first image and a second display area being maintained in a turn-off state during a hybrid mode; a source driving circuit configured to supply a first source signal to first pixels disposed in the first display area and supply a second source signal to second pixels disposed in the second display area to sense electrical characteristics of the second pixels when the display panel operates in the hybrid mode; and a timing controller configured to transfer first image data for the first image and sensing data for sensing to the source driving circuit and compensate for second image data for a second image to be displayed after displaying of the first image, based on a target compensation value obtained

from the electrical characteristics sensed by the source driving circuit, wherein the target compensation value is determined based on a third image displayed in a previous display mode of the hybrid mode, and when a vehicle driving start signal and a display panel raising signal are received from an electronic control unit (ECU) of the vehicle, the timing controller sets an operation mode of the display apparatus to the hybrid mode.

13. The vehicle of claim 12, wherein the timing controller comprises: a content type determiner configured to determine a content type of the third image displayed on the first display area and the second display area during the previous mode; and a data compensator configured to determine the target compensation value based on a compensation value mapped to the content type and compensate for the second image data by using the target compensation value, and wherein the second image data is image data for the second image to be displayed on the first display area and the second display area during a display mode to be performed after the hybrid mode.

14. The vehicle of claim 13, wherein luminance of the first image is lower than luminance of the third image.

15. The vehicle of claim 13, wherein, when luminance of the first image is higher than luminance of the third image, the data compensator compensates for the second image data by using a compensation value determined based on a sensing result obtained through a sensing mode performed after an operation of the hybrid mode.

16. The vehicle of claim 13, wherein the timing controller further comprises a first artificial neural network model trained using a training set including sensing values obtained in outputting image data corresponding to each content type for each content type, wherein the content type determiner inputs the sensing result to the first artificial neural network model to determine a result value, having a highest probability value among result values fed back from the first artificial neural network model, as a content type mapped to the sensing result.

17. The vehicle of claim 13, wherein the content type comprises a normal type including images for movie or drama, a graphics type including images for a user interface (UI) or a vehicular cluster, a sports type including a sports image, and a game type including a game image.

18. The vehicle of claim 12, wherein a gate driving circuit configured to sequentially supply a gate pulse to first gate lines disposed in the first display area to drive the first gate lines and supply the gate pulse to more than one target gate lines among second gate lines disposed in the second display area to drive the target gate lines, and wherein a plurality of continuous first gate lines are disposed in the first display area and a plurality of continuous second gate lines are disposed in the second display area.

19. The vehicle of claim 18, wherein the target gate lines are changed at every frame period.

20. The vehicle of claim 18, wherein, in operating in a display mode, the display panel displays an image on the first display area and the second display area, the source driving circuit supplies a source signal for displaying the image to the first and second pixels with, and the gate driving circuit sequentially supplies the gate pulse to gate lines disposed in the first and second display areas to sequentially drive the gate lines.
